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Chapter 1
Differential Calculus of Functions of Several Variables

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In the study of engineering, we rarely encounter physical quantities that depend on only one variable. The temperature of a building may depend on its position and time; the stress in a mechanical component may depend on several spatial coordinates; the voltage and current in an electrical system may depend on time, position, and other circuit parameters; and the displacement of a structure may depend on its location and the loads applied to it. Thus, to understand and model many real engineering systems, we need mathematical functions that depend on several variables. In single-variable calculus, we study functions of the form

$$y = f(x),$$

where a change in one independent variable produces a change in the dependent variable. In engineering applications, however, we often encounter functions of the form

$$z = f(x, y)$$

or

$$w = f(x, y, z),$$

where several independent variables influence the quantity being studied. Differential calculus of several variables provides the mathematical tools for describing and analyzing these relationships. For example, consider the temperature T at different locations inside a room. The temperature may depend on the horizontal coordinates x and y, and therefore can be represented by

$$T = T(x, y).$$

If we want to know how rapidly the temperature changes as we move in the x-direction while keeping y fixed, we use a *partial derivative*. Similarly, the rate of change in the y-direction is described by another partial derivative. These ideas are fundamental in understanding heat distribution, thermal systems, fluid flow, and many other engineering problems.

For **electrical and computer engineering students**, functions of several variables arise naturally in the analysis of electric and magnetic fields, electrical potentials, signal behavior, semiconductor devices, communication systems, and electromagnetic waves. A field quantity may vary with position, for example,

$$V = V(x, y, z),$$

where V represents electric potential. The derivatives of such a function provide information about how the potential changes from one point to another and lead naturally to important engineering concepts such as electric fields and gradients.

For **mechanical engineering students**, multivariable calculus is essential for describing quantities such as displacement, velocity, temperature, pressure, stress, strain, and energy. For example, the temperature of a mechanical component may vary throughout its three-dimensional structure:

$$T = T(x, y, z).$$

Partial derivatives can then describe the rate at which temperature changes in different directions. Similarly, optimization techniques can be used to determine dimensions or operating conditions that minimize material, energy consumption, or manufacturing cost.

For **civil engineering students**, structures and construction materials are influenced by many variables simultaneously. The deflection of a beam, the stress within a structural member, the pressure in a fluid, and the distribution of temperature within a structure can all depend on several variables. Multivariable calculus provides tools for determining rates of change, approximating engineering quantities, and finding maximum and minimum values subject to physical constraints.

For **electromechanical engineering students**, the interaction between electrical and mechanical systems makes multivariable calculus particularly important. Position, velocity, current, voltage, force, torque, temperature, and electromagnetic quantities may influence one another. The chain rule, partial derivatives, gradients, and optimization methods provide a mathematical language for describing these coupled systems.

One of the most important ideas in this chapter is that a function of several variables can change in different directions at different rates. The *gradient*

$$\nabla f$$

collects the partial derivatives of a function into a vector and provides information about the direction of greatest increase of the function. This concept has applications ranging from electromagnetic fields and heat transfer to optimization, robotics, control systems, and computer engineering.

The idea of a *tangent plane* extends the familiar tangent line from single-variable calculus to surfaces. If

$$z = f(x, y),$$

then near a suitable point, the surface can be approximated by a plane. This gives engineers a practical method for estimating the change in an engineering quantity when the input variables

undergo small changes. Such approximations are particularly useful because exact mathematical models can often be complicated, while a local linear approximation may be sufficiently accurate for engineering purposes.

Another important application is *optimization*. Engineers frequently need to find the largest or smallest possible value of a quantity. For example, one may want to maximize the efficiency of a machine, minimize the cost of a structure, minimize energy consumption, maximize the strength-to-weight ratio of a component, or determine the dimensions of a design subject to certain restrictions. The concepts of relative extrema, absolute extrema, and *Lagrange multipliers* provide systematic mathematical methods for solving such problems.

Therefore, differential calculus of functions of several variables is not simply an extension of techniques learned in single-variable calculus. It is a mathematical framework for understanding how several interacting variables influence engineering systems. The ideas developed in this chapter—limits and continuity, partial derivatives, higher-order derivatives, directional derivatives,

gradients, total differentials, tangent planes, the chain rule, extrema, and constrained optimization—form a foundation for many advanced engineering courses.
As you study this chapter, try not to view the formulas merely as computational rules. Instead, ask an engineering question whenever you encounter a new concept:

> ***What does this derivative measure? In which direction is the quantity changing? What does the gradient tell us? How can a tangent plane approximate a real engineering system? What quantity should an engineer maximize or minimize?***

These questions will help connect the mathematics to the engineering problems for which it is ultimately intended.

- **Learning Outcomes**

By the end of this chapter, you should be able to:

1. describe and visualize functions of several variables using graphs and level curves;

2. understand limits and continuity of multivariable functions;
3. calculate and interpret partial derivatives;
4. calculate higher-order partial derivatives;
5. determine directional derivatives and gradients;
6. use total differentials to describe small changes in engineering quantities;
7. construct tangent planes and use them for approximation;
8. apply the chain rule and implicit differentiation;
9. determine relative maximum and minimum values;
10. determine largest and smallest values on specified sets; and
11. solve constrained optimization problems using Lagrange multipliers.

The ultimate goal is to develop the ability to **translate an engineering problem into a mathematical model, analyze how the model changes, and use the resulting information to make meaningful engineering decisions**.

DEFINITIONS AND EXAMPLES OF FUNCTIONS OF TWO VARIABLES

Definition(FUNCTIONS OF TWO VARIABLES)

A function f of two variables is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes on, that is, $\{f(x, y) | (x, y) \in D\}$

Example 1.1

For each of the following functions, evaluate $f(3,2)$ and find the domain.

1. $f(x,y) = \dfrac{\sqrt{x+y+1}}{x-1}$
2. $f(x,y) = x\ln(y^2 - x)$

Definition

If f is a function of two variables with domain D, then the graph of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and (x, y) is in D.

Example 1.2

Sketch the graph of the function

1. $f(x, y) = 6 - 3x - 2y$.
2. $g(x, y) = \sqrt{9 - x^2 - y^2}$.

LEVEL CURVE

DEFINITION

The level curves of a function f of two variables are the curves with equations $f(x, y) = k$, where k is a constant (in the range of f).

Example 1.3

Sketch the level curves of the function

1. $f(x, y) = 6 - 3x - 2y$ for $k = -6, 0, 6, 12$.
2. $g(x, y) = \sqrt{9 - x^2 - y^2}$ for $k = 0, 1, 2, 3$.

DEFINITIONS AND EXAMPLES OF FUNCTIONS OF THREE VARIABLES

DEFINITION

A function of three variables, f , is a rule that assigns to each ordered triple (x, y, z) in a domain $D \subset \mathbb{R}^3$ a unique real number denoted by $f(x, y, z)$. For instance, the temperature T at a point on the surface of the earth depends on the longitude x and latitude y of the point and on the time t, so we could write $T = f(x, y, t)$.

Example 1.4

Find the domain of f if $f(x, y, z) = \ln(z - y) + xy \sin(z)$

LEVEL SURFACES

DEFINITION

The level surfaces of a function f of three variables are the surfaces with equations $f(x, y, z) = k$, where k is a constant (in the range of f).

Example 1.5

Find the level surfaces of the function $f(x, y, z) = x^2 + y^2 + z^2$

DEFINITION

A function of n variables is a rule that assigns a number $z = f(x_1, x_2, \cdots, x_n)$ to an n -tuple $(x_1, x_2, \cdots, x_n)$ of real numbers. We denote by $\mathbb{R}^n$ the set of all such n-tuples.
For example, if a company uses n different ingredients in making a food product, c_i is the cost per unit of the ith ingredient, and x_i units of the ith ingredient are used, then the total cost C of the ingredients is a function of the n variables $x_1, x_2, \cdots, x_n$:

$$C = f(x_1, x_2, \cdots, x_n) = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

DEFINITION

Let f be a function of two variables whose domain D includes points arbitrarily close to (a, b) . Then we say that the limit of $f(x, y)$ as (x, y) approaches (a, b) is L and we write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

if for every number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that

$$if\ (x, y) \in\ \ D\ and\ 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$$

then

$$|f(x, y) - L| < \epsilon$$

For functions of a single variable, when we let x approach a , there are only two possible directions of approach, from the left or from the right. We recall from applied mathematics I that

$$\text{if } \lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x), \text{ then } \lim_{x \to a} f(x) \text{ does not exist}$$

For functions of two variables the situation is not as simple because we can let (x, y) approach (a, b) from an infinite number of directions in any manner as long as (x, y) stays within the domain of f .

The above definition says that the distance between $f(x, y)$ and L can be made arbitrarily small by making the distance from (x, y) to (a, b) sufficiently small (but not 0). The definition refers only to the distance between (x, y) and (a, b) . It does not refer to the direction of approach. Therefore, if the limit exists, then $f(x, y)$ must approach the same limit no matter how (x, y) approaches (a, b) . Thus if we can find two different paths of approach along which the function $f(x, y)$ has different limits, then it follows that $lim_{(x,y)\to(a,b)} f(x, y)$ doesnot exist.

DEFINITION

If $f(x, y) \to L_1$ as $(x, y) \to (a, b)$ along a path C_1 and $f(x, y) \to L_2$ as $(x, y) \to (a, b)$ along a path C_2, where $L_1 \neq L_2$, then $\lim\limits_{(x,y)\to(a,b)} f(x, y)$ does not exist.

Example 1.6

Show that $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Example 1.7

If $f(x,y) = \dfrac{xy}{x^2+y^2}$, does $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exist?.

Example 1.8

If $f(x,y) = \dfrac{xy^2}{x^2+y^4}$, does $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ exist?.

Example 1.9

Find $\lim_{(x,y)\to(0,0)} \dfrac{3x^2y}{x^2+y^2}$, if it exists?.

CONTINUITY OF FUNCTIONS OF TWO VARIABLES

Recall that evaluating limits of continuous functions of a single variable is easy. It can be accomplished by direct substitution because the defining property of a continuous function is $lim_{x \to a} f(x) = f(a)$. Continuous functions of two variables are also defined by the direct substitution property.

DEFINITION

A function f of two variables is called continuous at (a, b) if

$$\lim_{(x,y) \to (a,b)} f(x, y) = f(a, b)$$

We say f is continuous on D if f is continuous at every point (a, b) in D.

The intuitive meaning of continuity is that if the point (x, y) changes by a small amount, then the value of $f(x, y)$ changes by a small amount. This means that a surface that is the graph of a continuous function has no hole or break.

Using the properties of limits, you can see that sums, differences, products, and quotients of continuous functions are continuous on their domains. Let's use this fact to give examples of continuous functions.

A polynomial function of two variables (or polynomial, for short) is a sum of terms of the form $cx^m y^n$, where c is a constant and m and n are nonnegative integers. A rational function is a ratio of polynomials. For instance,

$$f(x, y) = x^4 + 5x^3y^2 + 6xy^4 - 7y + 6$$

is a polynomial, whereas

$$f(x, y) = \frac{2xy + 1}{x^2 + y^2}$$

is a rational function.

Example 1.10

Evaluate $\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y)$

SOLUTION.

Since $f(x, y) = x^2y^3 - x^3y^2 + 3x + 2y$ is a polynomial, it is continuous everywhere, so we can find the limit by direct substitution:

$$\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y) = 1^2.2^3 - 1^3.2^2 + 3(1) + 2(2) = 11$$

Example 1.11

Where is the function $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$ continuous?.

SOLUTION.

The function f is discontinuous at $(0, 0)$ because it is not defined there. Since f is a rational function, it is continuous on its domain, which is the set

$$D = \{(x, y) | (x, y) \neq (0, 0)\}$$

.

Example 1.12

Let

$$g(x,y) = \begin{cases} \dfrac{x^2 - y^2}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Here g is defined at $(0,0)$ but g is still discontinuous there because $lim_{(x,y)\to(0,0)} g(x,y)$ does not exist.

Example 1.13

Let

$$g(x,y) = \begin{cases} \dfrac{3x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

We know f is continuous for $(x,y) \neq (0,0)$ since it is equal to a rational function there. Also, we have

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0 = f(0,0)$$

Therefore f is continuous at $(0,0)$, and so it is continuous on $\mathbb{R}^2$.

Just as for functions of one variable, composition is another way of combining two continuous functions to get a third. In fact, it can be shown that if f is a continuous function of two variables and g is a continuous function of a single variable that is defined on the range of f , then the composite function $h = g \circ f$ defined by $h(x, y) = g(f(x, y))$ is also a continuous function.

Example 1.14

Where is the function $f(x, y) = \arctan(y/x)$ continuous?.

SOLUTION.

The function $f(x, y) = y/x$ is a rational function and therefore continuous except on the line $x = 0$. The function $g(t) = \arctan(t)$ is continuous everywhere. So the composite function

$$g(f(x, y)) = \arctan(y/x) = h(x, y)$$

is continuous except where $x = 0$.

Note

Everything that we have done in functions of two variables can be extended to functions of three or more variables.

DEFINITION

If f is a function of two variables, its partial derivatives are the functions f_x and f_y defined by

$$f_x(x,y) = \lim_{h \to 0} \frac{f(x+h,y) - f(x,y)}{h}$$

$$f_y(x,y) = \lim_{h \to 0} \frac{f(x,y+h) - f(x,y)}{h}$$

NOTATIONS FOR PARTIAL DERIVATIVES

If $z = f(x, y)$, we write

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

RULE FOR FINDING PARTIAL DERIVATIVES OF $z = f(x, y)$

1. To find f_x , regard y as a constant and differentiate $f(x, y)$ with respect to x.
2. To find f_y , regard x as a constant and differentiate $f(x, y)$ with respect to y

Example 1.15

If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ and $f_y(2, 1)$.

SOLUTION.

Holding y constant and differentiating with respect to x , we get

$$f_x(x, y) = 3x^2 + 2xy^3 \Rightarrow f_x(2, 1) = 16$$

and

Holding x constant and differentiating with respect to y, we get

$$f_y(x, y) = 3x^2y^2 - 4y \Rightarrow f_y(2, 1) = 8$$

Example 1.16

If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

SOLUTION.

Using the Chain Rule for functions of one variable, we have

$$\frac{\partial f}{\partial x} = \cos\left(\frac{x}{1+y}\right).\frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right).\frac{1}{1+y}$$

$$\frac{\partial f}{\partial y} = \cos\left(\frac{x}{1+y}\right).\frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = -\cos\left(\frac{x}{1+y}\right).\frac{x}{(1+y)^2}$$

PARTIAL DERIVATIVES VIEWED AS RATES OF CHANGE AND SLOPES

Recall that if $y = f(x)$, then the value of $f'(x_0)$ can be interpreted either as the rate of change of y with respect to x at the point x_0 or as the slope of the tangent line to the graph of f at the point x_0. Partial derivatives have analogous interpretations. To see that this is so, suppose that C_1 is the intersection of the surface $z = f(x, y)$ with the plane $y = y_0$ and that C_2 is its intersection with the plane $x = x_0$.

Thus, $f_x(x, y_0)$ can be interpreted as the rate of change of z with respect to x along the curve C_1 and $f_y(x_0, y)$ can be interpreted as the rate of change of z with respect to y along the curve C_2.

PARTIAL DERIVATIVES VIEWED AS RATES OF CHANGE AND SLOPES

In particular $f_x(x_0, y_0)$ is the rate of change of z with respect to x along the curve C_1 at the point (x_0, y_0) and $f_y(x_0, y_0)$ is the rate of change of z with respect to y along the curve C_2 at the point (x_0, y_0). Geometrically, $f_x(x_0, y_0)$ can be viewed as the slope of the tangent line to the curve C_1 at the point (x_0, y_0), and $f_y(x_0, y_0)$ can be viewed as the slope of the tangent line to the curve C_2 at the point (x_0, y_0). We will call $f_x(x_0, y_0)$ the slope of the surface in the x-direction at (x_0, y_0), and $f_y(x_0, y_0)$ the slope of the surface in the y-direction at (x_0, y_0).

Example 1.17

Let $f(x, y) = x^2y + 5y^3$

1. Find the slope of the surface $z = f(x, y)$ in the x- direction at the point $(1, -2)$.
2. Find the slope of the surface $z = f(x, y)$ in the y- direction at the point $(1, -2)$.

SOLUTION.

1. Differentiating f with respect to x with y held fixed yields

$$f_x = 2xy$$

Thus, the slope in the x-direction is $f_x(1, -2) = -4$; that is, z is decreasing at the rate of 4 units per unit in x.

2. Differentiating f with respect to y with x held fixed yields

$$f_y = x^2 + 15y^2$$

Thus, the slope in the y-direction is $f_y(1, -2) = 61$; that is, z is increasing at the rate of 61 units per unit increase in y.

Example 1.18

Find the slope of the sphere, $x^2 + y^2 + z^2 = 1$ in the y-direction at the points $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$ and $(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3})$.

SOLUTION.

The point $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$ lies on the upper hemisphere

$$z = \sqrt{1 - x^2 - y^2},$$

and the point $(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3})$ lies on the lower hemisphere

$$z = -\sqrt{1 - x^2 - y^2}.$$

We could find the slopes by differentiating each expression for z separately with respect to y and then evaluating the derivatives at $x = \frac{2}{3}$ and $y = \frac{1}{3}$. However, it is more efficient to differentiate the given equation

$$x^2 + y^2 + z^2 = 1$$

implicitly with respect to y since this will give us both slopes with one differentiation.

To perform the implicit differentiation, we view z as a function of x and y and differentiate both sides with respect to y, taking x to be fixed. The computations are as follows:

$$\frac{\partial}{\partial y}(x^2 + y^2 + z^2) = \frac{\partial}{\partial y}(1)$$

$$\Rightarrow 0 + 2y + 2z\frac{\partial z}{\partial y} = 0 \Rightarrow \frac{\partial z}{\partial y} = -\frac{y}{z}$$

Substituting the y- and z-coordinates of the points $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$ and $(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3})$ in this expression, we find that the slope at the point $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$ is $-\frac{1}{2}$ and the slope at $(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3})$ is $\frac{1}{2}$.

Since the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$, are functions of x and y, these functions may themselves have partial derivatives. This gives rise to four possible second-order partial derivatives of f, which are defined by

$$\underbrace{\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = f_{xx}}_{\text{Differentiate twice with respect to x}}$$

$$\underbrace{\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = f_{yy}}_{\text{Differentiate twice with respect to y}}$$

$$\underbrace{\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = f_{xy}}$$

Differentiate first with respect to x and then with respect to y

$$\underbrace{\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = f_{yx}}$$

Differentiate first with respect to y and then with respect to x

The last two cases are called the mixed second-order partial derivatives or the mixed second partials. Also, the derivatives $\partial f/\partial x$ and $\partial f/\partial y$ are often called the first-order partial derivatives when it is necessary to distinguish them from higher-order partial derivatives.

Example 1.19

Find the second-order partial derivatives of $f(x, y) = x^2y^3 + x^4y$.

SOLUTION.

We have

$$\frac{\partial f}{\partial x} = 2xy^3 + 4x^3y \text{ and } \frac{\partial f}{\partial y} = 3x^2y^2 + x^4$$

so that

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(2xy^3 + 4x^3y) = 2y^3 + 12x^2y$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}(3x^2y^2 + x^4) = 6x^2y$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(3x^2y^2 + x^4) = 6xy^2 + 4x^3$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}(2xy^3 + 4x^3y) = 6xy^2 + 4x^3$$

HIGHER ORDER PARTIAL DERIVATIVES

Third-order, fourth-order, and higher-order partial derivatives can be obtained by successive differentiation. Some possibilities are

$$\frac{\partial^3 f}{\partial x^3} = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial x^2}\right) = f_{xxx} \qquad \frac{\partial^4 f}{\partial y^4} = \frac{\partial}{\partial y}\left(\frac{\partial^3 f}{\partial y^3}\right) = f_{yyyy}$$

$$\frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial y \partial x}\right) = f_{xyy} \qquad \frac{\partial^4 f}{\partial y^2 \partial x^2} = \frac{\partial}{\partial y}\left(\frac{\partial^3 f}{\partial y \partial x^2}\right) = f_{xxyy}$$

Example 1.20

Let $f(x, y) = y^2 e^x + y$. Find f_{xyy}

SOLUTION.

$$f_{xyy} = \frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial^2}{\partial y^2}\left(\frac{\partial f}{\partial x}\right)$$

$$= \frac{\partial^2}{\partial y^2}(y^2 e^x) = \frac{\partial}{\partial y}(2ye^x) = 2e^x$$

PARTIAL DERIVATIVES OF FUNCTIONS WITH MORE THAN TWO VARIABLES

For a function $f(x, y, z)$ of three variables there are three partial derivatives

$$f_x(x, y, z),\ f_y(x, y, z),\ f_z(x, y, z)$$

The partial derivative, $f_x(x, y, z)$ is calculated by holding y and z constant and differentiating with respect to x. For f_y the variables x and z are held constant, and for f_z the variables x and y are held constant. If a dependent variable

$$w = f(x, y, z)$$

is used, then the three partial derivatives of f can be denoted by

$$\frac{\partial w}{\partial x},\ \frac{\partial w}{\partial y},\ \text{and}\ \frac{\partial w}{\partial z}$$

Example 1.21

If $f(x,y,z) = x^3y^2z^4 + 2xy + z$,then

$f_x = 3x^2y^2z^4 + 2y$

$f_y = 2x^3yz^4 + 2x$

$f_z = 4x^3y^2z^3 + 1$

$f_z(-1,1,2) = 4(-1)^3(1)^2(2)^3 + 1 = -31.$

Example 1.22

If $f(p, \theta, \phi) = p^2 \cos(\phi) \sin(\theta)$, then

$f_p(p, \theta, \phi) = 2p \cos(\phi) \sin(\theta)$

$f_{p\phi}(p, \theta, \phi) = -2p \sin(\phi) \sin(\theta)$

$f_{p\phi\theta}(p, \theta, \phi) = -2p \sin(\phi) \cos(\theta)$

PARTIAL DERIVATIVES OF FUNCTIONS WITH MORE THAN TWO VARIABLES

In general, if $f(v_1, v_2, \cdots, v_n)$ is a function of n variables, there are n partial derivatives of f, each of which is obtained by holding $(n-1)$ of the variables fixed and differentiating the function f with respect to the remaining variable. If $w = f(v_1, v_2, \cdots, v_n)$, then these partial derivatives are denoted by

$$\frac{\partial w}{\partial v_1}, \frac{\partial w}{\partial v_2}, \cdots, \frac{\partial w}{\partial v_n}$$

where $\partial w/\partial v_i$ is obtained by holding all variables except v_i fixed and differentiating with respect to v_i.

Example 1.23

Find

$$\frac{\partial}{\partial x_i}\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

for $i = 1, 2, \cdots, n$.

SOLUTION.

For each $i = 1, 2, \cdots, n$ we obtain

$$\frac{\partial}{\partial x_i}\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

$$= \frac{1}{2\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}} \cdot \frac{\partial}{\partial x_i}[x_1^2 + x_2^2 + \cdots + x_i^2 + \cdots + x_n^2]$$

$$= \frac{1}{2\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}} . 2x_i$$

$$= \frac{x_i}{\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}}$$

TANGENT PLANE

DEFINITION

Suppose f has continuous partial derivatives. An equation of the tangent plane to the surface $z = f(x, y)$ at the point $P_0 = (x_0, y_0, z_0)$ is

$$z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Example 1.24

Find the tangent plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

SOLUTION. Let $f(x, y) = 2x^2 + y^2$. Then

$$f_x = 4x \Rightarrow f_x(1, 1) = 4, \text{ and } f_y = 2y \Rightarrow f_y(1, 1) = 2$$

Thus the equation of the tangent plane at $(1, 1, 3)$ is

$$z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

or

$$z = 3 + f_x(1, 1)(x - 1) + f_y(1, 1)(y - 1)$$

$$\Rightarrow z = 3 + 4(x - 1) + 2(y - 1) \Rightarrow z = 4x + 2y - 3$$

Example 1.25

Find an equation for the tangent plane and parametric equations for the normal line to the surface $z = x^2 y$ at the point $(2, 1, 4)$.

SOLUTION. Since $f(x,y) = x^2 y$, it follows that

$$f_x(x,y) = 2xy, \text{ and } f_y(x,y) = x^2$$

so that, with $x = 2$, $y = 1$,

$$f_x(2,1) = 4, \text{ and } f_y(2,1) = 4$$

Thus, a vector normal to the surface at $(2,1,4)$ is

$$n = f_x(2,1)i + f_y(2,1)j - k = 4i + 4j - k$$

Therefore, the tangent plane has the equation

$$4(x-2) + 4(y-1) - (z-4) = 0 \text{ or } 4x + 4y - z = 8$$

and the normal line has equations

$$x = 2 + 4t, \; y = 1 + 4t, \; z = 4 - t$$

Recall that a function f of one variable is called differentiable at x_0 if it has a derivative at x_0, that is, if the limit

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

exists.

A function f that is differentiable at a point x_0 has two important properries:

1. $f(x)$ is continuous at x_0.
2. The graph of $y = f(x)$ has a nonvertical tangent lines at x_0.

Our primary objective in this section is to extend the notion of differentiability to functions of two variables in such a way that the natural analogs of these two properties hold. More precisely, when $f(x, y)$ is differentiable at (x_0, y_0), we will want it to be the case that

1. $f(x, y)$ is continuous at (x_0, y_0).
2. The surface $z = f(x, y)$ has a nonvertical tangent plane at (x_0, y_0).

It would not be unreasonable to conjecture that a function f of two variables should be called differentiable at (x_0, y_0) if the two partial derivatives $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist. Unfortunately, this condition is not strong enough to meet our objectives, since there are functions that have partial derivatives at a point but are not continuous at that point. For example, consider the function

$$f(x, y) = \begin{cases} -1 & \text{if } x > 0 \text{ and } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

This function is discontinuous at $(0, 0)$ but has partial derivatives at $(0, 0)$; these derivatives are $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$(why?).

To motivate an appropriate definition of differentiability for functions of two variables, it will be helpful to reexamine the definition of differentiability for a function f of one variable. To say that f is differentiable at x_0 means that there exists a number $f'(x_0)$, which we call the derivative of f at x_0, such that

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \qquad [*]$$

For convenience. Let us express the numerator as

$$\Delta f = f(x_0 + \Delta x) - f(x_0)$$

which allows us to rewrite $[*]$ more compactly as

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} \Rightarrow \lim_{\Delta x \to 0} \left[\frac{\Delta f}{\Delta x} - f'(x_0) \right] = 0 \qquad (*)$$

Now, let us define ϵ [or more accurately $\epsilon(\Delta x)$], to be the error in approximating $f'(x_0)$ by $\Delta f/\Delta x$; that is,

$$\epsilon = \frac{\Delta f}{\Delta x} - f'(x_0)$$

Thus, we can rewrite $(*)$ as

$$\lim_{\Delta x \to 0} \epsilon = 0$$

which suggests the following altenative definition of differentiability for functions of one variable.

DEFINITION

A function f of one variable is said to be differentiable at x_0 if there exists a number $f'(x_0)$ such that Δf can be written in the form

$$\Delta f = f'(x_0)\Delta x + \epsilon \Delta x$$

where ϵ is a function of Δx such that $\epsilon \to 0$ as $\Delta x \to 0$, and $\epsilon = 0$ if $\Delta x = 0$.

Although this definition of differentiability is more complicated than that given in Applied mathematics I, it provides the basis for extending the notion of differentiability to functions of two or more variables.

If $z = f(x, y)$, then the change in the value of $f(x, y)$ as (x, y) moves from some initial position (x_0, y_0) to some new position $(x_0 + \Delta x, y_0 + \Delta y)$ is called the increment in f or the increment in z and is denoted by Δf or Δz. Thus,

$$\Delta f = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

we now make the following definition of differentiability for functions of two variables.

DEFINITION

A function f of two variables is said to be differentiable at (x_0, y_0) if $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist and Δf can be written in the form

$$\Delta f = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y$$

where ϵ_1 and ϵ_2 are functions of Δx and Δy such that $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$, and $\epsilon_1 = 0 = \epsilon_2$ if $(\Delta x, \Delta y) = (0, 0)$.

If a function f of two variables is differentiable at each point of a region $\mathbb{R}$ in the xy-plane, then we say that f is differentiable on $\mathbb{R}$; and if f is differentiable at every point in the xy-plane, then we say that f is differentiable everywhere. Moreover, we will say that f is a differentiable function if it is differentiable at each point of its domain.

THEOREM(SUFFICIENT CONDITIONS FOR DIFFERENTIABILITY)

If f has first order partial derivatives at each point in some circular region centered at (x_0, y_0), and if these partial derivatives are continuous at (x_0, y_0), then f is differentiable at (x_0, y_0).

Example 1.26

Show that $f(x, y) = x^3 y^4$ is a differentiable function.

SOLUTION.

The partial derivatives $f_x = 3x^2y^4$ and $f_y = 4x^3y^3$ are defined and continuous everywhere in the xy-p1ane.

Thus the hypotheses of the above Theorem(sufficient conditions for differentiability) are satisfied at each point (x_0, y_0) in the xy-plane, so $f(x, y) = x^3y^4$ is everywhere differentiable.

Theorem 1.27

If $f(x, y)$ is differentiable at (x_0, y_0), then f is continuous at (x_0, y_0).

RELATIONSHIP BETWEEN DIFFERENTIABILITY AND CONTINUITY

COROLLARY

If f has first-order partial derivatives at each point of some circular region centered at (x_0, y_0), and if this partial derivatives are continuous at (x_0, y_0), then f is continuous at (x_0, y_0).

Theorem 1.28

Let f be a function of two variables x and y. If f_x, f_y, f_{xy}, and f_{yx} are continuous on an open set, then $f_{xy} = f_{yx}$ at each point of the set.

Theorem

If $x = x(t)$ and $y = y(t)$ are differentiable at t, and
if $z = f(x, y)$ is differentiable at $(x(t), y(t))$, then $z = f(x(t), y(t))$ is differentiable at t, and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

Example 1.29

If $z = x^2y$, $x = t^2$, $y = t^3$,
then use the chain rule to find $\dfrac{dz}{dt}$, and check the result by expressing z as a function of t and differentiating directly.

SOLUTION. By the chain rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$= 2xy(2t) + x^2(3t^2)$$

$$= (2t^5)(2t) + (t^4)(3t^2) = 7t^6$$

Alternatively, we can express z directly as a function of t,

$$z = x^2y = (t^2)^2(t^3) = t^7$$

and then differentiate to obtain $dz/dt = 7t^6$. However, this procedure may not always be convenient.

Example 1.30

If $z = \sqrt{xy + y}$, $x = \cos(\theta)$, $y = \sin(\theta)$,
then use the chain rule to find $\dfrac{dz}{d\theta}$ when $\theta = \dfrac{\pi}{2}$.

SOLUTION. From the chain rule with θ in place of t,

$$\frac{dz}{d\theta} = \frac{\partial z}{\partial x}\frac{dx}{d\theta} + \frac{\partial z}{\partial y}\frac{dy}{d\theta}$$

we obtain

$$\frac{dz}{d\theta} = \frac{1}{2}(xy+y)^{-\frac{1}{2}}(y)(-\sin(\theta)) + \frac{1}{2}(xy+y)^{-\frac{1}{2}}(x+1)(\cos(\theta))$$

when $\theta = \dfrac{\pi}{2}$, we have

$$x = \cos(\pi/2) = 0 \text{ and } y = \sin(\pi/2) = 1$$

Substituting $x = 0, y = 1,$ and $\theta = \pi/2$ in the formula for $\dfrac{dz}{d\theta}$ yields

$$\left.\frac{dz}{d\theta}\right|_{\theta=\pi/2} = \frac{1}{2}(1)(1)(-1) + \frac{1}{2}(1)(1)(0) = -\frac{1}{2}$$

REMARK.

There are many variations in derivative notations, each of which gives the chain rule a different look.If $z = f(x, y)$, where x and y are functions of t, then some possibilities are

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$$

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

$$\frac{df}{dt} = f_x x'(t) + f_y y'(t)$$

In the special case where, $z = F(x, y)$ and y is a differentiable function of x, chain rule formula yields

$$\frac{dz}{dx} = \frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial y}\frac{dy}{dx}$$

$$= \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} \qquad [*]$$

This result can be used to find derivatives of functions that are defined implicitly. Suppose that the equation

$$F(x, y) = 0$$

defines y implicitly as a differentiable function of x and we are interested in finding dy/dx. Differentiating both sides of $F(x, y) = 0$ with respect to x and applying $[*]$ yields

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0$$

Thus, if $\partial F/\partial y \neq 0$, we obtain

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}$$

In summary, we have the following result.

Theorem

If the equation, $F(x, y) = 0$ defines y implicitly as a differentiable function of x, and if $\partial F/\partial y \neq 0$, then

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y} \qquad (*)$$

Example 1.31

Given that

$$x^3 + y^2 x - 3 = 0$$

find dy/dx using $(*)$, and check the result using implicit differentiation.

SOLUTION. By $(*)$ with $F(x, y) = x^3 + y^2x - 3$

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y} = -\frac{3x^2 + y^2}{2xy}$$

Alternatively, differentiating the given equation implicitly yields

$$3x^2 + y^2 + x\left(2y\frac{dy}{dx}\right) - 0 = 0 \Rightarrow \frac{dy}{dx} = -\frac{3x^2 + y^2}{2xy}$$

which agrees with the result obtained by $(*)$.

Theorem

If $x = x(u, v)$ and $y = y(u, v)$ have first-order partial derivatives at the point (u, v), and if $z = f(x, y)$ is differentiable at the point $(x(u, v), y(u, v))$, then $z = f(x(u, v), y(u, v))$ has first-order partial derivatives at (u, v) given by

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$

and

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

Example 1.32

Given that

$$z = e^{xy},\ x = 2u + v,\ y = u/v$$

find $\partial z/\partial u$ and $\partial z/\partial v$ using the chain rule.

SOLUTION.

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} = (ye^{xy})(2) + (xe^{xy})\left(\frac{1}{v}\right)$$

$$= \left[2y + \frac{x}{v}\right]e^{xy} = \left[\frac{2u}{v} + \frac{2u+v}{v}\right]e^{(2u+v)(u/v)}$$

$$= \left[\frac{4u}{v} + 1\right]e^{(2u+v)(u/v)}$$

and

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v} = (ye^{xy})(1) + (xe^{xy})\left(-\frac{u}{v^2}\right)$$

$$= \left[y - x\left(\frac{u}{v^2}\right)\right]e^{xy} = \left[\frac{u}{v} - (2u+v)\frac{u}{v^2}\right]e^{(2u+v)(u/v)}$$

$$= -\frac{2u^2}{v^2}e^{(2u+v)(u/v)}$$

Example 1.33

At what rate is the area of a rectangle changing if its length is 8 ft and increasing at 3 ft/s while its width is 6 ft and increasing at 2 ft/s?.

SOLUTION. Let
x =length of the rectangle in feet.
y = width of the rectangle in feet.
A = area of the rectangle in square feet.
t = time in seconds.
We are given that

$$\frac{dx}{dt} = 3 \text{ and } \frac{dy}{dt} = 2 \qquad [*]$$

at the instant when

$$x = 8 \text{ and } y = 6 \qquad (*)$$

We want to find dA/dt at the instant. For this purpose we use the area formula $A = xy$ to obtain

$$\frac{dA}{dt} = \frac{\partial A}{\partial x}\frac{dx}{dt} + \frac{\partial A}{\partial y}\frac{dy}{dt} = y\frac{dx}{dt} + x\frac{dy}{dt}$$

Substituting $(*)$ and $[*]$ in this equation yields

$$\frac{dA}{dt} = 6(3) + 8(2) = 34$$

Thus, the area is increasing at a rate of $34\ ft^2/s$ at the given instant.

Now we suppose that z is given implicitly as a function $z = f(x, y)$ by an equation of the form $F(x, y, z) = 0$. This means that $F(x, y, f(x, y)) = 0$ for all (x, y) in the domain of f . If F and f are differentiable, then we can use the Chain Rule to differentiate the equation $F(x, y, z) = 0$ as follows:

$$\frac{\partial F}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial F}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x} = 0$$

but

$$\frac{\partial x}{\partial x} = 1 \text{ and } \frac{\partial y}{\partial x} = 0$$

so this equation becomes

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z} \text{ provided that } \frac{\partial F}{\partial z} \neq 0$$

The formula for $\partial z/\partial y$ is obtained in a similar manner, that is

$$\frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z} \text{ provided that } \frac{\partial F}{\partial z} \neq 0$$

Example 1.34

Find $\partial z/\partial x$ and $\partial z/\partial y$ if $x^3 + y^3 + z^3 + 6xyz = 1$

SOLUTION. Let $F(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1 = 0$ then

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{3x^2 + 6yz}{3z^2 + 6xy} = -\frac{x^2 + 2yz}{z^2 + 2xy}$$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{3y^2 + 6xz}{3z^2 + 6xy} = -\frac{y^2 + 2xz}{z^2 + 2xy}$$

TOTAL DIFFERENTIALS OF FUNCTIONS OF TWO VARIABLES

Recall that if $y = f(x)$ is a function of one variable, then the differential

$$dy = f'(x_0)dx$$

represents the change in y along the tangent line at (x_0, y_0) produced by a change dx in x and

$$\Delta y = f(x_0 + \Delta x) - f(x_0)$$

represents the change in y along the curye $y = f(x)$ produced by a change dx in x.

Analogously, if $z = f(x, y)$ is a function of two variables, we will define dz to be the change in z along the tangent plane at (x_0, y_0, z_0) and Δz to the surface $z = f(x, y)$ produced by changes dx and dy in x and y respectively. This is in contrast to

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \qquad [1]$$

which represents the change in z along the surface produced by changes dx and dy in x and y.

TOTAL DIFFERENTIALS OF FUNCTIONS OF TWO VARIABLES

To derive a formula for dz, $p_0(x_0, y_0, z_0)$ be a fixed point on the surface $z = f(x, y)$. If we assume f to be differentiable at (x_0, y_0), then the surface has a tangent plane at p_0, given by the equation

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + z - z_0 = 0$$

or

$$z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \qquad [2]$$

It follows from [2] that the tangent plane has height z_0 when $x = x_0$, $y = y_0$, and it has height

$$z_0 + f_x(x_0, y_0)dx + f_y(x_0, y_0)dy \qquad [3]$$

when $x = x_0 + dx$, $y = y_0 + dy$. Thus, the change dz in the height of the tangent plane as (x, y) varies from (x_0, y_0) to $(x_0 + dx, y_0 + dy)$ is obtained by subtracting z_0 from expression [3]. This yields

$$dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy \qquad [4]$$

Often, we will omit the subscripts on x_0 and y_0 in formulas [1] and [4] and write these equations as

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y) \qquad [5]$$

$$dz = f_x(x, y)dx + f_y(x, y)dy \qquad [6]$$

We call dz the total differential of z or the total differential of f. In these formulas, Δx, Δy, dx, and dy are usually viewed as variables and x and y as fixed numbers. When it is desirable not to use a dependent variable, we will write df for the total differential of f.

Example 1.35

1. If $z = f(x, y) = x^2 + 3xy - y^2$, find the total differential dz.
2. If x changes from 2 to 2.05 and y changes from 3 to 2.96 , compare the values of Δz and dz .

SOLUTION.

1. $dz = \dfrac{\partial z}{\partial x}dx + \dfrac{\partial z}{\partial y}dy = (2x + 3y)dx + (3x - 2y)dy.$
2. Putting $x = 2$, $dx = \Delta x = 0.05$, $y = 3$, and $dy = \Delta y = -0.04$, we get

$$dz = [(2)(2) + (3)(3)]0.05 + [3(2) - 2(3)](-0.04) = 0.65$$

The increment of z is

$$\Delta z = f(2.05, 2.96) - f(2, 3)$$

$$= [(2.05)^2 + 2(3)(2.05) - (2.96)^2] - [2^2 + 3(2)(3) - 3^2]$$

$$= 0.6449$$

Notice that $\Delta z \approx dz$ but dz is easier to compute.

Example 1.36

The base radius and height of a right circular cone are measured as 10 cm and 25 cm, respectively, with a possible error in measurement of as much as 0.1 cm in each. Use differentials to estimate the maximum error in the calculated volume of the cone.

SOLUTION.

The volume V of a cone with base radius r and height h is $V = \dfrac{\pi r^2 h}{3}$. So the differential of V is

$$dV = \frac{\partial V}{\partial r}dr + \frac{\partial V}{\partial h}dh = \frac{2\pi rh}{3}dr + \frac{\pi r^2}{3}dh$$

Since each error is at most 0.1 cm, we have $|\Delta r| \leq 0.1$,$|\Delta h| \leq 0.1$. To find the largest error in the volume we take the largest error in the measurement of r and of h . Therefore we take $dr = 0.1$ and $dh = 0.1$ along with $r = 10$, $h = 25$. This gives

$$dV = \frac{500\pi}{3}(0.1) + \frac{100\pi}{3}(0.1) = 20\pi$$

Thus the maximum error in the calculated volume is about $20\pi cm^3 \approx 63cm^3$.

We know that an equation of the tangent plane to the graph of the function $f(x, y) = 2x^2 + y^2$ at the point $(1, 1, 3)$ is

$$z = 4x + 2y - 3.$$

Therefore, the linear function of two variables

$$L(x, y) = 4x + 2y - 3$$

is a good approximation to $f(x, y)$ when (x, y) is near $(1, 1)$.

The function L is called the linearization of f at $(1,1)$ and the approximation

$$f(x,y) \approx 4x + 2y - 3$$

is called the linear approximation or tangent plane approximation of f at $(1,1)$. For instane, at the point $(1.1, 0.95)$ the linear approximation gives

$$f(1.1, 0.95) \approx 4(1.1) + 2(0.95) - 3 = 3.3$$

which is quite close to the true value of $f(1.1, 0.95) = 2(1.1)^2 + 0.95^2 = 3.3225$. But if we take a point farther away from $(1,1)$, such as $(2,3)$, we no longer get a good approximation. In fact, $L(2,3) = 11$ whereas $f(2,3) = 17$.

In general, we know that an equation of the tangent plane to the graph of a function f of two variables at the point $(a, b, f(a, b))$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

The linear function whose graph is this tangent plane, namely

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the linearization of f at (a, b) and the approximation

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the linear approximation or the tangent plane approximation of f at (a, b).

THEOREM

If the partial derivatives f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b).

Example 1.37

Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$ and find its linearization there. Then use it to approximate $f(1.1, -0.1)$.

SOLUTION.
The partial derivatives are

$$f_x(x,y) = e^{xy} + xye^{xy},\ f_y(x,y) = x^2e^{xy}$$

$$f_x(1,0) = 1,\ f_y(1,0) = 1$$

Both f_x and f_y are continuous functions, so f is differentiable by the above Theorem. The linearization is

$$L(x,y) = f(1,0) + f_x(1,0)(x-1) + f_y(1,0)(y-0) = x + y$$

The corresponding linear approximation is

$$xe^{xy} \approx x + y,\ \text{so } f(1.1,-0.1) \approx 1.1 + (-0.1) = 1$$

Compare this with the actual value of $f(1.1,-0.1) = 1.1e^{(-0.1)(1.1)} = 0.98542$.

The partial derivatives $f_x(x, y)$ and $f_y(x, y)$ represent the rates of change of $f(x, y)$ in the directions parallel to the x- and y-axis. In this section we will investigate rates of change of $f(x, y)$ in other directions.

Our first objective is to determine how to find the slope of a surface $z = f(x, y)$ at a point (x_0, y_0) in an arbitrary specified direction. Since there are infinitely many directions in which a point can move in the xy-plane, we need some method for describing a specific direction starting at (x_0, y_0).

One way to do this is to use a unit vector

$$u = u_1 i + u_2 j$$

that has its initial point at (x_0, y_0) and points in the desired direction. This vector determines a line l in the xy-plane that can be expressed parametrically as

$$x = x_0 + su_1 \qquad y = y_0 + su_2 \qquad [*]$$

where s is the parameter that has its reference point at (x_0, y_0) and has positive values in the direction of u. For $s = 0$, the point (x, y) is at the reference point (x_0, y_0) and it moves along the line l in the direction of u as s increases. We will call the rate of change of z with respect to s along the surface $z = f(x, y)$ at $s = 0$ the slope of the surface in the direction u at (x_0, y_0).

If f is differentiable at (x_0, y_0), then this slope can be calculated by finding the value of dz/ds at $s = 0$. To do this, we start with the chain rule

$$\frac{dz}{ds} = f_x(x, y)\frac{dx}{ds} + f_y(x, y)\frac{dy}{ds}$$

$$\Rightarrow \frac{dz}{ds} = f_x(x, y)u_1 + f_y(x, y)u_2 \text{ from } [*]$$

at $s = 0$, the point (x, y) is (x_0, y_0), so

$$\left.\frac{dz}{ds}\right|_{s=0} = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2 \qquad (*)$$

which is a formula for the slope of the surface in the direction of u at (x_0, y_0).

If we now keep the vector $u = u_1 i + u_2 j$ in Formula $(*)$ fixed but replace the specific point (x_0, y_0) by a general point (x, y), then the resulting expression

$$f_x(x, y)u_1 + f_y(x, y)u_2$$

is a function of x and y whose value at the point (x, y) is the slope of the surface in the direction of u at that point. This "slope-producing function" is sufficiently important that it has some terminology and notation associated with it.

DEFINITION

If f is a differentiable function of x and y, and if $u = u_1 i + u_2 j$ is a unit vector, then the directional derivative of f in the direction of u is denoted by $D_u f$ and defined by

$$D_u f(x, y) = f_x(x, y)u_1 + f_y(x, y)u_2$$

Example 1.38

Find the directional derivative of $f(x, y) = 3x^2y$ at the point $(1, 2)$ in the direction of the vector $a = 3i + 4j$.

SOLUTION. The vector a is not a unit vector, so we must normalize it to apply the Formula. This yields

$$u = \frac{a}{||a||} = \frac{1}{\sqrt{25}}(3i + 4j) = \frac{3}{5}i + \frac{4}{5}j$$

from which we obtain $u_1 = \frac{3}{5}$ and $u_2 = \frac{4}{5}$. Since the partial derivatives of f are

$$f_x = 6xy, \ f_y = 3x^2$$

Thus the directional derivative of f in the direction of u is

$$D_u f(x, y) = f_x(x, y)u_1 + f_y(x, y)u_2 = 6xy\left(\frac{3}{5}\right) + 3x^2\left(\frac{4}{5}\right)$$

Thus,

$$D_u f(1,2) = \frac{36}{5} + \frac{12}{5} = \frac{48}{5}$$

Note that the value of $D_u f(x, y)$ depends on the point (x, y) and the direction of u. Thus, at a fixed point the slope of the surface may vary with the direction. Since a unit vector u can be expressed as

$$u = \cos(\phi)i + \sin(\phi)j$$

where ϕ is the angle from the positive x-axis to u. Thus, the directional derivative of f in the direction of u can also be expressed as

$$D_u f(x, y) = f_x(x, y)\cos(\phi) + f_y(x, y)\sin(\phi)$$

Example 1.39

Find the directional derivative of $f(x, y) = e^{xy}$ at $(-2, 0)$ in the direction of the unit vector that makes an angle of $\frac{\pi}{3}$ with the positive x-axis.

SOLUTION. The partial derivatives of f are

$$f_x = ye^{xy} \Rightarrow f_x(-2,0) = 0, \ f_y = xe^{xy} \Rightarrow f_y(-2,0) = -2$$

The unit vector that makes an angle of $\frac{\pi}{3}$ with the positive x-axis is

$$u = \cos\left(\frac{\pi}{3}\right)i + \sin\left(\frac{\pi}{3}\right)j$$

Thus

$$D_u f(-2,0) = f_x(-2,0)\cos\left(\frac{\pi}{3}\right) + f_y(-2,0)\sin\left(\frac{\pi}{3}\right) = -\sqrt{3}$$

It follows from the Formula that the directional derivative of a function f in the direction of the unit vector i is

$$D_i f(x, y) = f_x(x, y)\cos(0) + f_y(x, y)\sin(0) = f_x(x, y)$$

and the directional derivative of f in the direction of j is

$$D_j f(x, y) = f_x(x, y)\cos(\pi/2) + f_y(x, y)\sin(\pi/2) = f_y(x, y)$$

Thus, the partial derivative of f with respect to x is the directional derivative in the positive x-direction, and the partial derivative of f with respect to y is the directional derivative of f in the positive y- direction.

Since reversing the direction of a vector u reverses the signs of its components, it follows from the formula of directional derivative that

$$D_{-u}f(x,y) = -D_u f(x,y) \qquad [*]$$

Thus, for example,
$D_{-i}f(x,y) = -D_i f(x,y) = -f_x(x,y)$ and

$D_{-j}f(x,y) = -D_j f(x,y) = -f_y(x,y)$

Fomula $[*]$ makes sense intuitively if you view directional derivatives as slopes and consider the effect of walking in opposite directions on the side of a hill. At each point on the side of the hill the "uphill grade" and the "downhill grade" are the same, but in one direction you are ascending and in the opposite direction you are descending.

The directional derivative

$$D_u f(x, y) = f_x(x, y)u_1 + f_y(x, y)u_2$$

can be expressed in the form of a dot product as

$$D_u f(x, y) = (f_x(x, y)i + f_y(x, y)j).(u_1 i + u_2 j)$$

the second vector in the product is U. However, the first vector is new; it is called the gradient of f and is denoted by ∇f or $\nabla f(x, y)$. The notation ∇z is also used if $z = f(x, y)$.

DEFINITION

If f is a function of x and y, then the gradient of f is defined by

$$\nabla f(x, y) = f_x(x, y)i + f_y(x, y)j$$

Example 1.40

Find the gradient of $f(x, y) = 3x^2y$ at the point $(1, 2)$ and use it to calculate the directional derivative of f at $(1, 2)$ in the direction of the vector $a = 3i + 4j$.

SOLUTION. Since

$$\nabla f(x,y) = f_x(x,y)i + f_y(x,y)j = 6xyi + 3x^2 j$$

so the gradient of f at $(1,2)$ is

$$\nabla f(1,2) = 12i + 3j$$

The unit vector in the direction of a is

$$U = \frac{a}{||a||} = \frac{1}{5}(3i + 4j) = \frac{3}{5}i + \frac{4}{5}j$$

Thus

$$D_U f(1,2) = \nabla f(1,2).U = (12i + 3j).\left(\frac{3}{5}i + \frac{4}{5}j\right) = \frac{48}{5}$$

The gradient is not merely a notational device to simplify the formula for directional derivative: we will see that the length and direction of the gradient ∇f provide important information about the function f and the surface $z = f(x, y)$.

For example, suppose that $\nabla f \neq 0$, then

$$D_U f(x,y) = \nabla f(x,y).U = ||\nabla f(x,y)||||U|| \cos(\theta)$$

$$= ||\nabla f(x,y)|| \cos(\theta) \qquad (*)$$

where θ is the angle between $\nabla f(x,y)$ and U. This equation tells us that the maximum value of $D_U f(x,y)$ is $||\nabla f(x,y)||$, and this maximum occurs when $\theta = 0$, that is, when U is in the direction of $\nabla f(x,y)$. Geometrically, this means that the surface $z = f(x,y)$ has its maximum slope at a point (x,y) in the direction of the gradient, and the maximum slope is $||\nabla f(x,y)||$.

Similarly, $(*)$ tells us that the minimum value of $D_U f(x,y)$ is $-||\nabla f(x,y)||$, and this minimum occurs when $\theta = \pi$, that is, when U is oppositely directed to $\nabla f(x,y)$. Geometrically, this means that the surface $z = f(x,y)$ has its minimum slope at a point (x,y) in the direction that is opposite to the gradient, and the minimum slope is $-||\nabla f(x,y)||$.

Finally in the case where $\nabla f(x,y) = 0$, it follows from $(*)$ that $D_U f(x,y) = 0$ in all directions at the point (x,y). This typically occurs where the surface $z = f(x,y)$ has a "relative minimum", a "relative maximum", or a "saddle point".

THEOREM

Let f be a function of two variables that is differentiable at (x, y).

1. If $\nabla f(x, y) = 0$, then all directional derivative of f at (x, y) is zero.
2. If $\nabla f(x, y) \neq 0$, then among all possible directional derivatives of f at (x, y) the derivative in the direction of $\nabla f(x, y)$ has the largest value. The value of that directional derivative is $||\nabla f(x, y)||$.
3. If $\nabla f(x, y) \neq 0$, then among all possible directional derivatives of f at (x, y) the derivative in the direction opposite to that of $\nabla f(x, y)$ has the smallest value. The value of that directional derivative is $-||\nabla f(x, y)||$.

Example 1.41

Let $f(x, y) = x^2 e^y$ Find the maximum value of a directional derivative at (-2.0), and find the unit vector in the direction in which the maximum value occurs.

SOLUTION. Since

$$\nabla f(x,y) = f_x(x,y)i + f_y(x,y)j = 2xe^y i + x^2 e^y j$$

the gradient of f at $(-2,0)$ is

$$\nabla f(-2,0) = -4i + 4j$$

By the above Theorem, the maximum value of the directional derivative is

$$||\nabla f(-2,0)|| = \sqrt{(-4)^2 + (4)^2} = \sqrt{32} = 4\sqrt{2}$$

This maximum occurs in the direction of $\nabla f(-2,0)$. The unit vector in this direction is

$$\frac{\nabla f(-2,0)}{||\nabla f(-2,0)||} = \frac{1}{4\sqrt{2}}(-4i + 4j) = -\frac{1}{\sqrt{2}}i + \frac{1}{\sqrt{2}}j$$

THEOREM

If f is differentiable at (x_0, y_0) and if $\nabla f(x_0, y_0) \neq 0$, then $\nabla f(x_0, y_0)$ is normal to the level curve of f through (x_0, y_0).

REMARK.

If (x_0, y_0) is a point on the level curve $f(x, y) = c$, then the slope of the surface $z = f(x, y)$ at that point in the direction of U is

$$D_U f(x_0, y_0) = \nabla f(x_0, y_0).U$$

If U is tangent to the level curve at (x_0, y_0), then $f(x, y)$ is neither increasing nor decreasing in that direction, so $D_U f(x_0, y_0) = 0$. Thus, $\nabla f(x_0, y_0)$, $-\nabla f(x_0, y_0)$, and the tangent vector U mark the directions of maximum slope, minimum slope, zero slope at a point (x_0, y_0) on a level curve.

Example 1.42

Sketch the level curve for the function $f(x, y) = x^2 + y^2$ through the point $(3, 4)$, and draw the gradient vector at this point.

SOLUTION. Since $f(3,4) = 25$, the level curve through the point $(3,4)$ has the equation $f(x,y) = 25$, which is the circle

$$x^2 + y^2 = 25$$

Since

$$\nabla f(x,y) = f_x(x,y)i + f_y(x,y)j = 2xi + 2yj$$

the gradient vector at $(3,4)$ is

$$\nabla f(3,4) = 6i + 8j$$

Note that the gradient vector is perpendicular to the circle at $(3,4)$, as guaranteed by the above Theorem.

In applied mathematics I we learned how to find maximum and minimum values of functions of one variable. In this section we will develop similar techniques for functions of two variables.

DEFINITION.

1. A function f of two variables is said to have a relative maximum at a point (x_0, y_0) if there is a circle centered at (x_0, y_0) such that $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the domain of f that lie inside the circle, and f is said to have an absolute maximum at (x_0, y_0) if $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the domain of f.
2. A function f of two variables is said to have a relative minimum at a point (x_0, y_0) if there is a circle centered at (x_0, y_0) such that $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) in the domain of f that lie inside the circle, and f is said to have an absolute minimum at (x_0, y_0) if $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) in the domain of f.

If f has a relative maximum or a relative minimum at (x_0, y_0), then we say that f has a relative extremum at (x_0, y_0), and if f has absolute maximum or absolute minimum at (x_0, y_0), then we say that f has absolute extremum at (x_0, y_0) .

THEOREM(Extreme-Value Theorem)

If $f(x, y)$ is continuous on a closed and bounded set R, then f has both an absolute maximum and an absolute minimum on R.

Recall that if a function g of one variable has a relative extremum at a point x_0 where g is differentiable, then $g'(x_0) = 0$. To obtain the analog of this result for functions of two variables, suppose that $f(x, y)$ has a relative maximum at a point (x_0, y_0) and that the partial derivatives of $f(x, y)$ exist at a point (x_0, y_0). It seems plausible geometrically that the traces of the surface $z = f(x, y)$ on the planes $x = x_0$ and $y = y_0$ have horizontal tangent lines at (x_0, y_0), so

$$f_x(x_0, y_0) = 0 \text{ and } f_y(x_0, y_0) = 0$$

The same conclusion holds if f has a relative minimum at (x_0, y_0) all of which suggests the following result, which we state without formal proof.

THEOREM

If f has a relative extremum at a point (x_0, y_0), and if the first-order partial derivatives of f exist at this point, then

$$f_x(x_0, y_0) = 0 \text{ and } f_y(x_0, y_0) = 0$$

Recall that the critical points of a function f of one variable are those values of x at which $f'(x) = 0$ or f is not differentiable. The following definition is the analog for functions of two variables.

DEFINITION

A point (x_0, y_0) is called a critical point of the function $f(x, y)$ if $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$ or if one or both partial derivatives does not exist at (x_0, y_0).

For functions of one variable the second derivative test was used to determine the behavior of a function at a critical point. The following theorem, which is usually proved in advanced calculus, is the analog for functions of two variables.

THEOREM(The Second Partial Test)

Let f be a function of two variables with continuous second order partial derivatives in some circle centered at a critical point (x_0, y_0), and let

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

1. If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a relative minimum at (x_0, y_0).
2. If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a relative maximum at (x_0, y_0).
3. If $D < 0$, then f has a saddle point at (x_0, y_0).
4. If $D = 0$, then no conclusion can be drawn.

Example 1.43

Locate all relative extrema and saddle points of

$$f(x, y) = 3x^2 - 2xy + y^2 - 8y$$

SOLUTION Since $f_x = 6x - 2y$ and $f_y = -2x + 2y - 8$, the critical points of f satisfy the equations

$$\begin{cases} 6x - 2y = 0 \\ -2x + 2y - 8 = 0 \end{cases}$$

Solving these for x and y yields $x = 2, y = 6(verify)$, so $(2, 6)$ is the only critical point. To apply the above Theorem we need the second-order partial derivatives

$$f_{xx} = 6,\ f_{yy} = 2,\ f_{xy} = -2$$

At the point $(2, 6)$ we have

$$D = f_{xx}(2,6)f_{yy}(2,6) - f_{xy}^2(2,6) = 6(2) - (-2)^2 = 8 > 0$$

$$\text{and } f_{xx}(2,6) = 6 > 0$$

So f has a relative minimum at $(2, 6)$ by part (1) of the second partials test.

Example 1.44

Locate all relative extrema and saddle points of

$$f(x, y) = 4xy - x^4 - y^4$$

SOLUTION. Since $f_x = 4y - 4x^3$ and $f_y = 4x - 4y^3$, the critical points of f satisfy the equations

$$\begin{cases} 4y - 4x^3 = 0 \\ 4x - 4y^3 = 0 \end{cases} \Leftrightarrow \begin{cases} y = x^3 \\ x = y^3 \end{cases} \qquad (*)$$

substituting the top equation in the bottom yields $x = (x^3)^3$ or $x^9 - x = 0$ or $x(x^8 - 1) = 0$, which has solutions $x = 0$, $x = 1$, and $x = -1$.

Substituting these values in the top equation of $(*)$, we obtain the corresponding y values $y = 0$, $y = 1$, $y = -1$. Thus, the critical points of f are $(0, 0)$, $(1, 1)$, and $(-1, -1)$.

Now, since

$$f_{xx} = -12x^2,\ f_{yy} = -12y^2,\ \text{and } f_{xy} = 4$$

which yields the following table:

Critical point (x_0, y_0)	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$	$D = f_{xx}f_{yy} - f_{xy}^2$
$(0, 0)$	0	0	4	-16
$(1, 1)$	-12	-12	4	128
$(-1, -1)$	-12	-12	4	128

At the points $(1, 1)$ and $(-1, -1)$, we have $D > 0$ and $f_{xx} < 0$, so relative maxima occur at these critical points. f has a saddle point at $(0, 0)$ since $D < 0$.

THEOREM

If a function f of two variables has an absolute extremum(either an absolute maximum or an absolute minimum) at an interior point of its domain, then this extremum occurs at a critical point.

If f is continuous on a closed and bounded set R, then the Extreme-Value Theorem guarantees the existence of an absolute maximum and an absolute minimum of f on R. These absolute extrema can occur either on the boundary of R or in the interior of R, but if an absolute extremum occurs in the interior, then it occurs at a critical point by the above Theorem.
Thus, we are led to the following procedure for finding absolute extrema:

How to Find the Absolute Extrema of a Continuous Function f of two Variables on a Closed and Bounded Set R

- STEP 1. Find the critical points of f that lie in the interior of R.
- STEP 2. Find all boundary points at which the absolute extrema can occur.
- STEP 3. Evaluate $f(x, y)$ at the points obtained in the preceding steps. The largest of these values is the absolute maximum and the smallest the absolute minimum.

Example 1.45

Find the absolute maximum and minimum values of

$$f(x, y) = 3xy - 6x - 3y + 7$$

on the closed triangular region R with vertices $(0, 0)$, $(3, 0)$ and $(0, 5)$.

SOLUTION. Since $f_x = 3y - 6$ and $f_y = 3x - 3$, so all critical points occur where

$$\begin{cases} 3y - 6 = 0 \\ 3x - 3 = 0 \end{cases}$$

Solving these equations yields $x = 1$ and $y = 2$, so $(1, 2)$ is the only critical point. By drawing the region R you can observe that this critical point is in the interior of R.

Next, we want to determine the location of the points on the boundary of R at which the absolute extrema might occur. The boundary of R consists of three line segments, each of which we will treat separately:

The Line segnent between $(0,0)$ and $(3,0)$:

On this line segment we have $y = 0$, so $f(x,y)$ simplifies to a function of the single variable x,

$$f(x,0) = u(x) = 3x(0) - 6x - 3(0) + 7 = -6x + 7,\ 0 \le x \le 3$$

This function has no critical points because $u'(x) = -6$ is non zero for all x. Thus the extreme values of $u(x)$ occur at the endpoints $x = 0$ and $x = 3$ which correspond to the points $(0,0)$ and $(3,0)$ of R.

The Line segnent between $(0,0)$ and $(0,5)$:

On this line segment we have $x = 0$, so $f(x,y)$ simplifies to a function of the single variable y,

$$f(0,y) = v(y) = 3(0)y - 6(0) - 3y + 7 = -3y + 7,\ 0 \le y \le 5$$

This function has no critical points because $v'(y) = -3$ is non zero for all y. Thus the extreme values of $v(x)$ occur at the endpoints $y = 0$ and $y = 5$ which correspond to the points $(0,0)$ and $(0,5)$ of R.

The Line segnent between $(3, 0)$ and $(0, 5)$:

In the xy-plane, an equation for this line segment is

$$y = -\frac{5}{3}x + 5,\ 0 \le x \le 3 \qquad [*]$$

, so $f(x, y)$ simplifies to a function of the single variable x,

$f(x, y) = f(x, -5/3x + 5) = w(x)$

$= 3(x)(-5/3x + 5) - 6(-5/3x + 5) - 3(-5/3x + 5) + 7$

$\Rightarrow w(x) = -5x^2 + 14x - 8,\ 0 \le x \le 3$

Since $w'(x) = -10x + 14$, the equation $w'(x) = 0$ yields $x = \frac{7}{5}$ as the only critical point of w.

Thus, the extreme values of w occur either at the critical point $\frac{7}{5}$ or at the endpoints $x = 0$ and $x = 3$.

The endpoints corresponds to the points $(0, 5)$ and $(3, 0)$ of R, and from $[*]$ the critical point corresponds to $(\frac{7}{5}, \frac{8}{3})$.

Finally, the following Table lists the values of $f(x, y)$ at the interior critical point and at the points on the boundary where an absolute extremum can occur. From the table we conclude that the absolute maximum value of f is $f(0, 0) = 7$ and the absolute minimum value is $f(3, 0) = -11$.

(x, y)	$(0,0)$	$(3,0)$	$(0,5)$	$(\frac{7}{5}, \frac{8}{3})$	$(1,2)$
$f(x, y)$	7	-11	-8	$\frac{9}{5}$	1

In this section we will study a powerful new method for maximizing or minimizing a function subject to constraints on the variables. This method will help us to solve certain optimization problems that are difficult or impossible to solve using the methods studied in the last section.

THEOREM(Constrained-Extremum Principle for Two Variables and One Constraint)

Let f and g be functions of two variables with continuous first-order partial derivatives on some open set containing the constraint curve $g(x, y) = 0$, and assume that $\nabla g \neq 0$ at any point on this curve. If f has a constrained relative extremum, then this extremum occurs at a point (x_0, y_0) on the costraint curve at which the gradient vectors $\nabla f(x_0, y_0)$ and $\nabla g(x_0, y_0)$ are parallel; that is, there is some number λ such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0)$$

Example 1.46

At what point or points on the circle $x^2 + y^2 = 1$ does $f(x, y) = xy$ have an absolute maximum, and what is that maximum?.

SOLUTION. Since the circle $x^2 + y^2 = 1$ is a closed and bounded set, and since $f(x, y) = xy$ is a continuous function, it follows from the Extreme-Value Theorem that f has an absolute maximum and an absolute minimum on the circle. To find these extrema, we will use Lagrange multipliers to find the constrained relative extrema, and then we will evaluate f at those relative extrema to find the absolute extrema.

We want to maximize $f(x, y) = xy$ subject to the constraint

$$g(x, y) = x^2 + y^2 - 1 = 0 \qquad [*]$$

First we will look for relative extrema. For this purpose we will need the gradients

$$\nabla f(x, y) = yi + xj \text{ and } \nabla g(x, y) = 2xi + 2yj$$

From the formula for $\nabla g(x, y)$ we see that $\nabla g = 0$ if and only if $x = 0$ and $y = 0$, so $\nabla g \neq 0$ at any point on the circle $x^2 + y^2 = 1$. Thus, at a constrained relative extremum we must have

$$\nabla f = \lambda \nabla g \Leftrightarrow yi + xj = \lambda(2xi + 2yj)$$

which is equivalent to the pair of equations

$$y = 2x\lambda \text{ and } x = 2y\lambda$$

It follows from these equations that if $x = 0$, then $y = 0$, and if $y = 0$, then $x = 0$. In either case we have $x^2 + y^2 = 0$, so the constraint equation $x^2 + y^2 = 1$ is not satisfied. Thus, we can assume that x and y are non zero, and we can rewrite the equations as

$$\lambda = \frac{y}{2x} \text{ and } \lambda = \frac{x}{2y}$$

from which we obtain

$$\frac{y}{2x} = \frac{x}{2y} \Leftrightarrow y^2 = x^2 \qquad (*)$$

substituting this in equation $[*]$ yields

$$2x^2 - 1 = 0$$

From which we obtain $x = \pm\frac{1}{\sqrt{2}}$. Each of these values when substituted in equation $(*)$, produces y-values of $y = \pm\frac{1}{\sqrt{2}}$. Thus, constrained relative extrema occur at the points $(1/\sqrt{2}, 1/\sqrt{2})$, $(1/\sqrt{2}, -1/\sqrt{2})$, $(-1/\sqrt{2}, 1/\sqrt{2})$, and $(-1/\sqrt{2}, -1/\sqrt{2})$.
The values of xy at these points are as follows:

(x, y)	$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$	$(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$	$(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$	$(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$
xy	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$

Thus, the function $f(x, y) = xy$ has an absolute maximum of $\frac{1}{2}$ occuring at the two points $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$.

Although it was not asked for, we can also see that f has an absolute minimum of $-\frac{1}{2}$ occuring at the two points
$(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$ and $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

REMARK

If c is a constant, then the functions $g(x, y)$ and $g(x, y) - c$ have the same gradient since the constant c drops out when we differentiate. Consequently, it is not essential to rewrite a constraint of the form $g(x, y) = c$ as $g(x, y) - c = 0$ in order to apply the constrained-extremum principle.

Thus, in the last example, we could have kept the constraint in the form $x^2 + y^2 = 1$ and then taken $g(x, y) = x^2 + y^2$ rather than $g(x, y) = x^2 + y^2 - 1$.

Example 1.47

Use the method of Lagrange multipliers to find the dimensions of a rectangle with perimeter p and maximum area.

SOLUTION. Let
x = length of the rectangle
y = width of the rectangle
A = area of the rectangle
We want to maximize $A = xy$ subject to the perimeter constraint

$$p = 2x + 2y \qquad [*]$$

If we let $f(x, y) = xy$ and $g(x, y) = 2x + 2y$, then we have

$$\nabla f = yi + xj \text{ and } \nabla g = 2i + 2j$$

Noting that $\nabla g \neq 0$, it follows from Lagrange's multiplier method that

$$yi + xj = \lambda(2i + 2j)$$

at a constrained relative maximum. This is equivalent to the two equations

$$y = 2\lambda \text{ and } x = 2\lambda$$

Eliminating λ from these equations we obtain $x = y$, which shows that the rectangle is actually a square. Using this condition and constraint $[*]$ we obtain, $x = \frac{p}{4}$ and $y = \frac{p}{4}$.

The method of Lagrange multipliers can also be used to maximize or minimize a function of three variables $f(x, y, z)$ subject to a constraint $g(x, y, z) = 0$. As a ru1e, the graph of $g(x, y, z) = 0$ will be some surface S in 3-space. Thus, from a geometric viewpoint, the problem is to maximize or minimize $f(x, y, z)$ as (x, y, z) varies over the surface S.

As usual, we distinguish between relative and absolute extrema. We will say that f has a constrained absolute maximum (minimum) at (x_0, y_0, z_0) if $f(x_0, y_0, z_0)$ is the largest (smallest) value of $f(x, y, z)$ on S, and we will say that f has a constrained relative maximum (minimum) at (x_0, y_0, z_0) if $f(x_0, y_0, z_0)$ is the largest (smallest) value of $f(x, y, z)$ on some patch of the surface S that contains (x_0, y_0, z_0) in its interior.

The following theorem, which we state without proof, is the three-variable analog of the

THEOREM(Constrained-Extremum Principle for Two Variables and One Constraint).

THEOREM(Constrained-Extremum Principle for Three Variables and One Constraint)

Let f and g be functions of three variables with continuous first-order partial derivatives on some open set containing the constraint surface $g(x, y, z) = 0$, and assume that $\nabla g \neq 0$ at any point on this surface. If f has a constrained relative extremum, then this extremum occurs at a point (x_0, y_0, z_0) on the costraint surface at which the gradient vectors $\nabla f(x_0, y_0, z_0)$ and $\nabla g(x_0, y_0, z_0)$ are parallel; that is, there is some number λ such that

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0)$$

Example 1.48

Find the points on the sphere

$$x^2 + y^2 + z^2 = 36$$

that are closest to and farthest from the point $(1, 2, 2)$.

SOLUTION. To avoid radicals, we will find points on the sphere that minimize and maximize the square of the distance to $(1, 2, 2)$. Thus, we want to find the relative extrema of

$$f(x, y, z) = (x - 1)^2 + (y - 2)^2 + (z - 2)^2$$

subject to the constraint

$$x^2 + y^2 + z^2 = 36 \qquad [*]$$

If we let $g(x, y, z) = x^2 + y^2 + z^2$, then $\nabla g = 2xi + 2yj + 2zk$. Thus, $\nabla g = 0$ if and only if $x = y = z = 0$. It follows that $\nabla g \neq 0$ at any point of the constraint surface $[*]$, and hence the constrained relative extrema must occur at points where

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$$

that is,

$$2(x-1)i + 2(y-2)j + 2(z-2)k = \lambda(2xi + 2yj + 2zk)$$

which leads to the equations

$$2(x-1) = 2x\lambda, \quad 2(y-2) = 2y\lambda, \quad 2(z-2) = 2z\lambda \qquad (*)$$

we may assume that x, y, and z are non zero since $x = 0$ doesn't satisfy the first equation, $y = 0$ doesn't satisfy the second, and $z = 0$ doesn't satisfy the third. Thus we can rewrite $(*)$ as

$$\frac{(x-1)}{x} = \lambda, \quad \frac{(y-2)}{y} = \lambda, \quad \frac{(z-2)}{z} = \lambda$$

The first two equation imply that

$$\frac{(x-1)}{x} = \frac{(y-2)}{y}$$

from which it follows that

$$y = 2x \qquad (**)$$

Similarly, the first and third equations imply that

$$z = 2x \qquad (***)$$

Substituting $(**)$ and $(***)$ in the constraint equation $[*]$, we obtain

$$9x^2 = 36 \Rightarrow x = \pm 2$$

Substituting these values in $(**)$ and $(***)$ yields two points

$$(2, 4, 4) \text{ and } (-2, -4, -4)$$

Since $f(2, 4, 4) = 9$ and $f(-2, -4, -4) = 81$, it follows that $(2, 4, 4)$ is the point on the sphere closest to $(1, 2, 2)$, and $(-2, -4, -4)$ is the point that is farthest.

Example 1.49

Use Lagrange multipliers to determine the dimensions of a rectangular box, open at the top, having a volume of $32 ft^3$, and requiring the least amount of material for its construction.

SOLUTION. Let
x = length of the box(in feet)
y = width of the box(in feet)
z = height of the box(in feet)
S = surface area of the box(in square feet)
The problem is to minimize the surface area

$$S = xy + 2xz + 2yz$$

subject to the volume constraint

$$xyz = 32 \qquad [*]$$

If we let $f(x, y, z) = xy + 2xz + 2yz$ and $g(x, y, z) = xyz$, then

$$\nabla f = (y + 2z)i + (x + 2z)j + (2x + 2y)k$$

and

$$\nabla g = (yz)i + (xz)j + (xy)k$$

It follows that $\nabla g \neq 0$ at any point on the surface $xyz = 32$, since x, y, and z are all nonzero on this surface. Thus, at a constrained relative extremum we must have $\nabla f = \lambda \nabla g$, that is

$$(y+2z)i+(x+2z)j+(2x+2y)k=\lambda((yz)i+(xz)j+(xy)k)$$

This condition yields the three equations

$$y+2z=\lambda yz,\ x+2z=\lambda xz,\ 2x+2y=\lambda xy$$

Because x, y, and z are nonzero, these equations can be rewritten as

$$\frac{1}{z}+\frac{2}{y}=\lambda,\ \frac{1}{z}+\frac{2}{x}=\lambda,\ \frac{2}{y}+\frac{2}{x}=\lambda$$

From the first two equations,

$$y = x \qquad (*)$$

and from the first and third equations,

$$z = \frac{1}{2}x \qquad (**)$$

Substituting $(*)$ and $(**)$ in the volume constraint $[*]$ yields

$$\frac{1}{2}x^3 = 32$$

This equation, together with $(*)$ and $(**)$, yields

$$x = 4, \quad y = 4, \quad z = 2$$

So the box using least material has a height of $2\,ft$ and a square base whose edges are $4\,ft$ long.